CLASS 09 / OCT 20 / U05

ATTRACTORS

P3 CONTINUES: THE RULE STOPS COMING FROM THE INDEX

MIDTERM UPLOAD CLOSED

P3c + P3study ISSUED TODAY

AN ATTRACTOR IS A NUMBER

For every panel: how far is it from something? Use that number to drive a parameter.

THE WHOLE IDEA

```
for each panel p:
    d = distance(p, a)
    value = f(d)
```

Everything else this week is a choice of f.

MEASURE IT FIRST

$$d = \sqrt{((px - ax)^2 + (py - ay)^2)}$$

Pythagoras, once per panel. The answer is in model units.

Index told you where in the sequence. Distance tells you how far from something outside the collection.

IN ARRAYS YOU BUILT THE POSITIONS. NOW YOU MEASURE THEM.

READ THE FIELD BEFORE YOU CHANGE ANYTHING

Distance is a value at every location. Look at it as grey before it becomes geometry.

01 MEASURE	02 DISPLAY	03 REMAP	04 DRIVE
Every location on the facade grid gets a number: how far to the attractor.	Show those numbers as grey. This is a diagram already.	Push that range into the range your parameter needs.	Only now does the panel change. The field was finished first.

THE FIELD, READ AS A GREY VALUE ACROSS THE FACADE · NEAR → FAR



IF THE GREY IMAGE IS NOT ALREADY A GOOD DIAGRAM, THE PANEL WILL NOT SAVE IT.

3.4 shows the attractors and the resulting variation side by side, with the parameter labelled. That order is the assignment.

REMAP, AND SAY WHERE YOU CLAMPED

Distance comes out in model units. Your parameter needs a specific range.

THE OPERATION

```
t   = (d − dmin) / (dmax − dmin)
t   = clamp(t, 0, 1)
out = outmin + t × (outmax − outmin)

d     0 … 4 units          ← input
out   0.85 … 0.15 opening  ← output
```

SUBTRACT. DIVIDE. MULTIPLY. ADD.

Nothing here is new. It is the arithmetic of a transformation, applied to a distance instead of a coordinate.

Clamping is a decision, not a safety net. Decide what happens past the end of your range and say so. Swap the two output values and the pattern inverts.

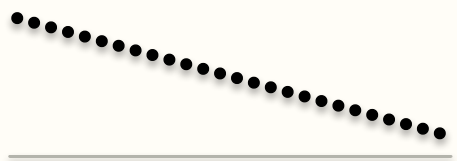
3.4 LABELS THE PARAMETER AND SAYS, IN PLAIN WORDS, HOW DISTANCE CHANGES IT.

FALLOFF, AND WHAT YOU MEASURE TO

f shapes the result. The attractor type changes only what the distance is measured to.

LINEAR

`v = 1 − t`



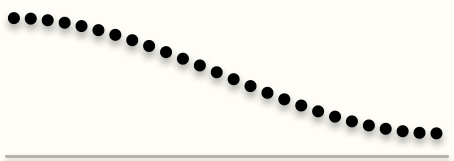
INVERSE

`v = 1 / (1 + k·t)`



SMOOTH

eased at both ends

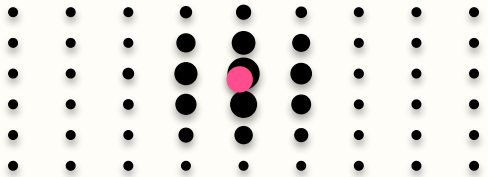


STEP

on / off at a cut

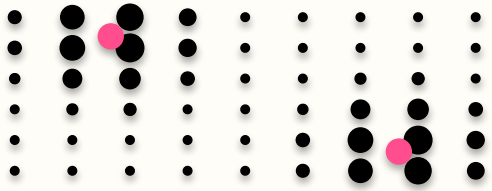


ONE POINT



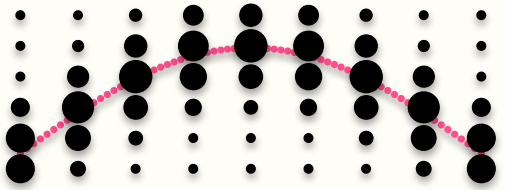
`d = distance(p, a)`

NEAREST OF SEVERAL



`d = min(distance(p, ai))`

NEAREST ON A CURVE



`d = distance(p, closest(c, p))`

SAME FACADE, SAME PANEL, SAME PARAMETER, THREE ATTRACTORS: 3.4 SHOWS WHAT THE ATTRACTOR ALONE CHANGES.

THE SAME THING, WRITTEN OUT

Measure, remap, clamp, shape the falloff, then drive one parameter. In that order.

```
# Pseudocode: choose ONE of the three distance methods.
for p in panel_locations:
    d = distance(p, point_a)                          # one point
    # d = min(distance(p, point_a), distance(p, point_b))
    # d = distance(p, closest_point(curve, p))

    opening = remap_clamped(d, from_range=(0, 4),
                            to_range=(0.85, 0.15))
    panel = build_panel(opening=opening)  # same prefab, new input
    place(panel, p)
```

WHAT IT MAPS ONTO

distance

external condition

field

remap

clamp

falloff

nearest

LINE 2 IS THE ONLY LINE THAT CHANGES BETWEEN THE THREE ATTRACTOR TYPES.

THE NUMBER DRIVES YOUR 3.1 PANEL

Not a sphere, not a cube. The panel you already built, rebuilt at every location with a different input.

01 MEASURE

Distance from each facade location to the attractor. Object Info, set to Relative, so the attractor and the facade share one coordinate system.

02 REMAP

One Map Range, Float, Linear, Clamp on. Input 0 to the reach of the attractor. Output: the safe range of your parameter, in its own units.

03 DRIVE

Instance on Points repeats a thing that is already built. To rebuild the panel per location with its own Opening, Depth or Height, the value goes into a For Each Element zone and reaches the panel's input inside it.

IF ALL PANELS LOOK THE SAME: THE VALUE ENTERED THE ZONE OUTSIDE, BUT NEVER REACHED THE INPUT INSIDE.

Attractors.blend, scenes 05 to 08: the supplied panel, then one point, several points and a curve driving its Opening.

ISSUED TODAY

P3study — ATTRACTORS - THREE FIELD STUDIES

DUE WEEK 10 — Oct 27

One facade, one panel, one controlled parameter, three attractors. Each diagram shows the attractor and the variation it produces, with the parameter labelled. No separate upload and no separate grade: it is page 1 of the 3.5 package.

DIAGRAM 1	DIAGRAM 2	DIAGRAM 3
ONE POINT Distance to a single point drives the parameter.	**SEVERAL POINTS** Say how they combine: nearest point, or an average.	**ONE CURVE** Distance to the nearest location along a curve.

ISSUED TODAY

P3c — ARCHITECTURAL APPLICATION

DUE WEEK 10 — Oct 27 / 10%

Reuse your 3.1 panel to design a simple facade, wall, canopy, pavilion or similar. Give it a purpose and a scale, choose the attractor that serves that purpose, and show why the variation matters architecturally.

PURPOSE	COMPARE	RESOLVE
WHAT IT DOES What the panel system encloses, screens, shades or defines.	**ONE CONTROLLED TEST** The attractor, the parameter it drives, and the difference it makes.	**ELEVATION + ISOMETRIC** Two sheets with 3.4: studies first, then the application.

REQUIREMENTS — P3c

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01	Reuse the panel you developed in 3.1 to design a simple facade, wall, canopy, pavilion or similar architectural application. You may develop your 3.2 architectural study further.
02	Give the design a clear purpose and scale. Explain what the panel system encloses, screens, shades or defines, and how attractor-driven variation supports that purpose.
03	Choose point, multiple-point or curve attractors. You do not need to combine every attractor type in the final design.
04	Show one controlled comparison and a resolved elevation and isometric. Identify the attractor, the geometric parameter it drives and why the variation matters architecturally.
05	A new precedent study is not required. Facadepot and other references may provide optional inspiration; credit any images or ideas you use.

NOW — YOUR FIRST FIELD

Rest of the class: measure, look, remap, and only then attach your panel.

✓	Distance to one point, measured across the facade grid you already have
✓	The field shown as grey on its own, before any panel changes
✓	Input and output ranges written down, in your parameter's units, with the clamp
✓	One parameter of your 3.1 panel driven: Opening, Depth, Height or similar. Just one
✓	The attractor moved, and the change predicted before you look

MOVE THE ATTRACTOR. IF YOU CANNOT PREDICT WHAT HAPPENS, STOP AND READ THE FIELD.

BEFORE NEXT CLASS

NEXT	Week 10 — Oct 27: 3.4 and 3.5 are submitted together at the start of class. Then 4.1 is introduced: the supplied field groups and a two-parameter matrix.
3.4	All three attractor types running on the same facade, same panel, same parameter. The comparison is what is being graded.
3.5	Pick the purpose first, then the attractor that serves it. One controlled comparison, one elevation, one isometric.
PANEL	Your 3.1 panel as a node group with the driven parameter exposed. Test two values by hand before you connect an attractor.